Mihir Mankikar

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Education

Texas A&M University, College Station, TX — Aug. 2024 – May 2028

Bachelor of Science in Computer Science, Minor in Statistics — GPA: 3.9

Engineering Honors; EH: Enhancing the Development of Global Engineers

Research

Automated Data Analysis for Defensive Cyber Operations (ADADCO) — Jan. 2026 – Present

- Collaborating with the U.S. Space Force to develop a machine learning-driven network intrusion dashboard for defensive cyber operations with real-time detection of DDoS, spoofing, and jamming threats
- Designing a condition monitoring system with operator alerts to visualize network health and streamline incident response

Calf Price Predictive Modeling in Cow-Calf Operations Using Univariate Modeling Frameworks for Decision Support — May 2025 – Present

- Developed a univariate time-series forecasting framework (ARIMA, SARIMA) for U.S. calf price prediction, implementing CPI-deflation to real prices and walk-forward validation with expanding windows
- Extended baseline models to SARIMAX by incorporating exogenous economic predictors as multivariate regressors, improving prediction accuracy
- Implemented rigorous model selection using grid search across parameter combinations with AIC/BIC criteria
- Designed automated forecasting pipeline with comprehensive performance metrics and residual diagnostics to support decision-making for long-term sustainability in cow-calf operations under economic uncertainty

Advancing Decision Support in Cow-Calf Operations Through Multivariate and Deep Learning-Based Prediction of Calf Prices — May 2025 – Present

- Engineered predictive features including temporal indicators, lag structures, rolling statistics, and differenced variables, with Boruta feature selection algorithm to identify statistically significant predictors
- Implemented machine learning pipelines comparing 4 approaches (Random Forest, AdaBoost, LightGBM, SVR) with Optuna-based hyperparameter optimization
- Developed deep learning architecture using LSTM networks with sequence modeling to capture non-linear temporal dependencies in price dynamics

Extracurriculars

Texas A&M Rocket Engine Design — Jan. 2026 – Present

Avionics and Controls Systems Member — *College Station, TX*

- Implemented binary communications protocol with automatic acknowledgment/retransmission logic and packet ID tracking for reliable UDP data exchange between Python GUI and embedded hardware
- Developed an automated abort system with 10ms heartbeat monitoring and engine state detection to prevent hardware damage during hot-fire tests

Ignitors Rocketry — Jan. 2026 – Present

Mentee — *College Station, TX*

- Completing a technical curriculum on rocketry fundamentals, including aerodynamic drag analysis, motor impulse classification, and recovery systems
- Performing OpenRocket flight simulations to analyze stability margins and predicted apogee
- Assembling and preparing three rockets for successful flight & recovery

Projects

Southwest Airlines Weather Impact Score | *Python, scikit-learn, PyTorch, Streamlit*

3D Kalman Filtering Target Tracker | *C++, Eigen, Python*

Celebrity-Style Insult Generator | *Python, Flask, aitextgen, NLTK, Beautiful Soup*

Skills

Programming Languages: Python, C++, Java, JavaScript, HTML/CSS, R
Data Science: TensorFlow, PyTorch, scikit-learn, statsmodels, NumPy, Pandas, LightGBM, XGBoost
Web Development: Django, Flask, React, React Native, Node.js, Angular, Firebase
Developer Tools: Git, VS Code, PyCharm, RStudio
Spoken Languages: English (Native), Marathi (Native), Japanese (Beginner/N5)